Modeling Report

Camas

Model date: July 2, 2022

State of the Snowpack

Interpretation Notes

Elevation bands are in increments of 1000 ft, e.g., 4000 is 4000 - 5000 ft.

Mean SWE is calculated over the total basin area. Mean cold content is calculated over the snow covered area to reflect the ripeness of the existing snowpack.

Surface Water Input (SWI) includes liquid water leaving the bottom of the snowpack and rain on bare ground.

Forcing data is created with the Spatial Modeling for Resources Framework (SMRF) using the High Resolution Rapid Refresh (HRRR) output fields as inputs.

Model Results

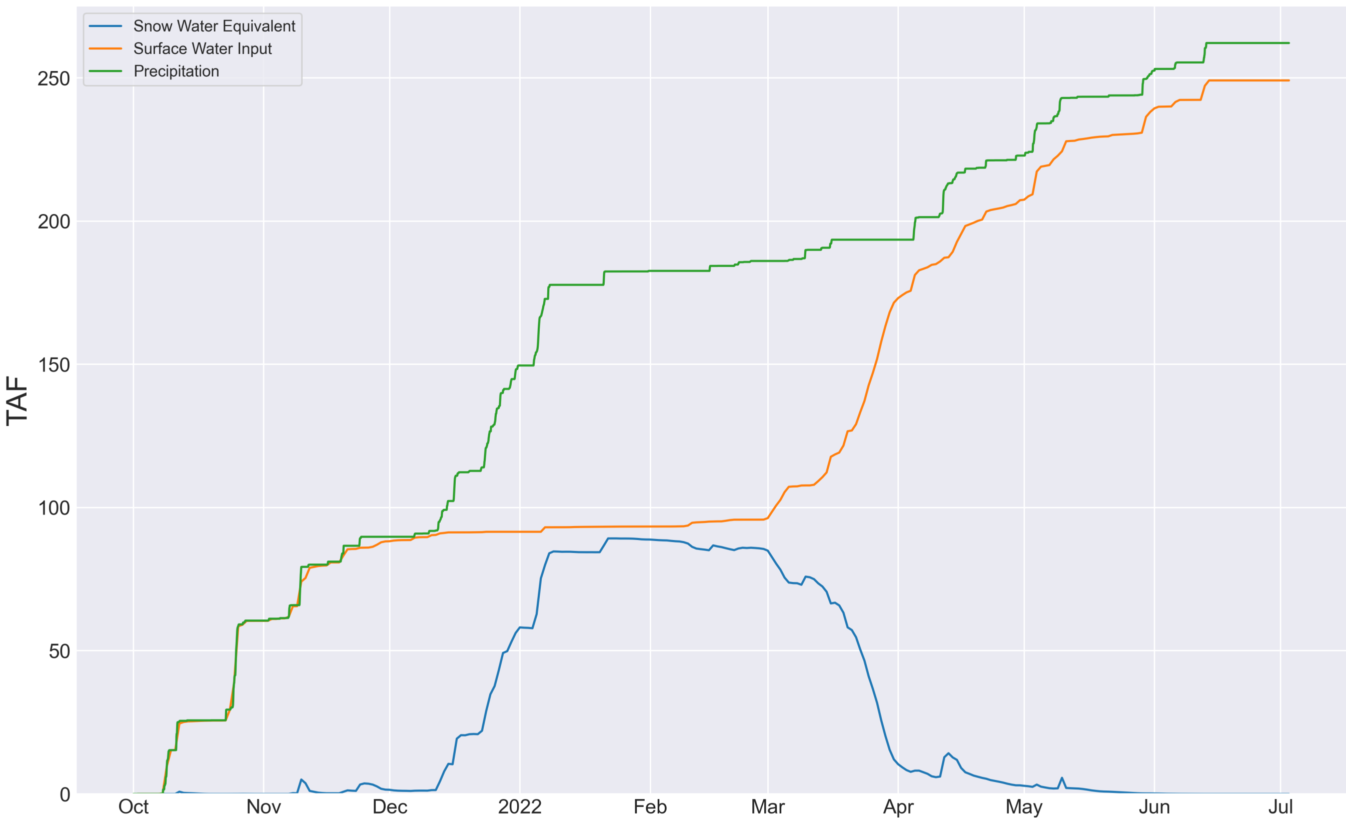

Fig.1 Water Year timeseries.

Model to station comparisons are a comparison of 50 meter cells to a point within the basin. Some differences are expected when comparing modeled and measured snowpack behavior at these different spatial scales. ASO 3m snow depth measurements at each station are indicated by the red '+' symbols. Absence of these symbols indicates that the station location or data quality are suspect, and they were not used in the ASO survey validation and report.

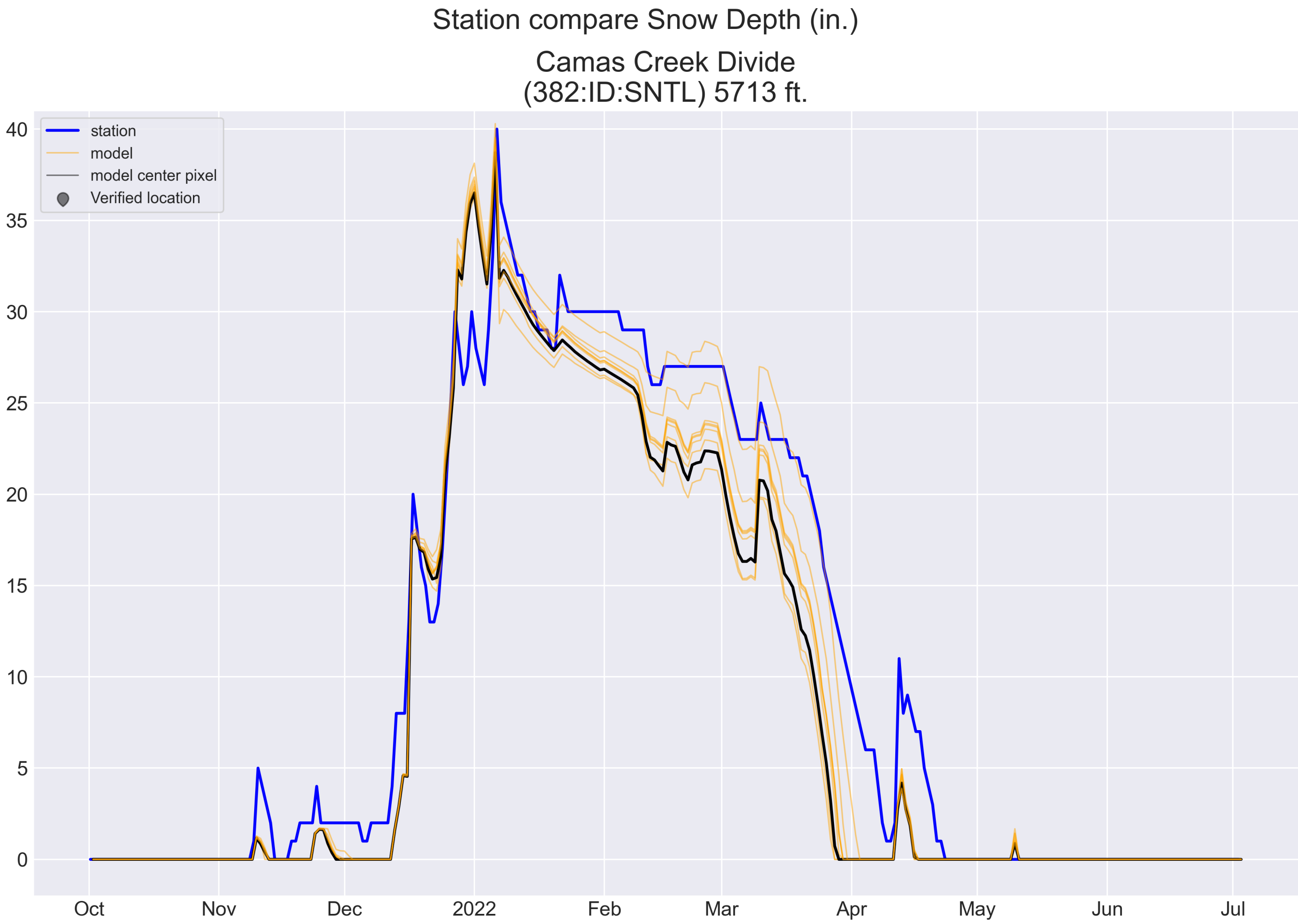

Fig.2 Comparison of modeled and measured Snow Depth [in.]. Comparisons of model pixels to measured station data are performed with the 9 nearest model pixels to give a more spatially complete view of model behavior.

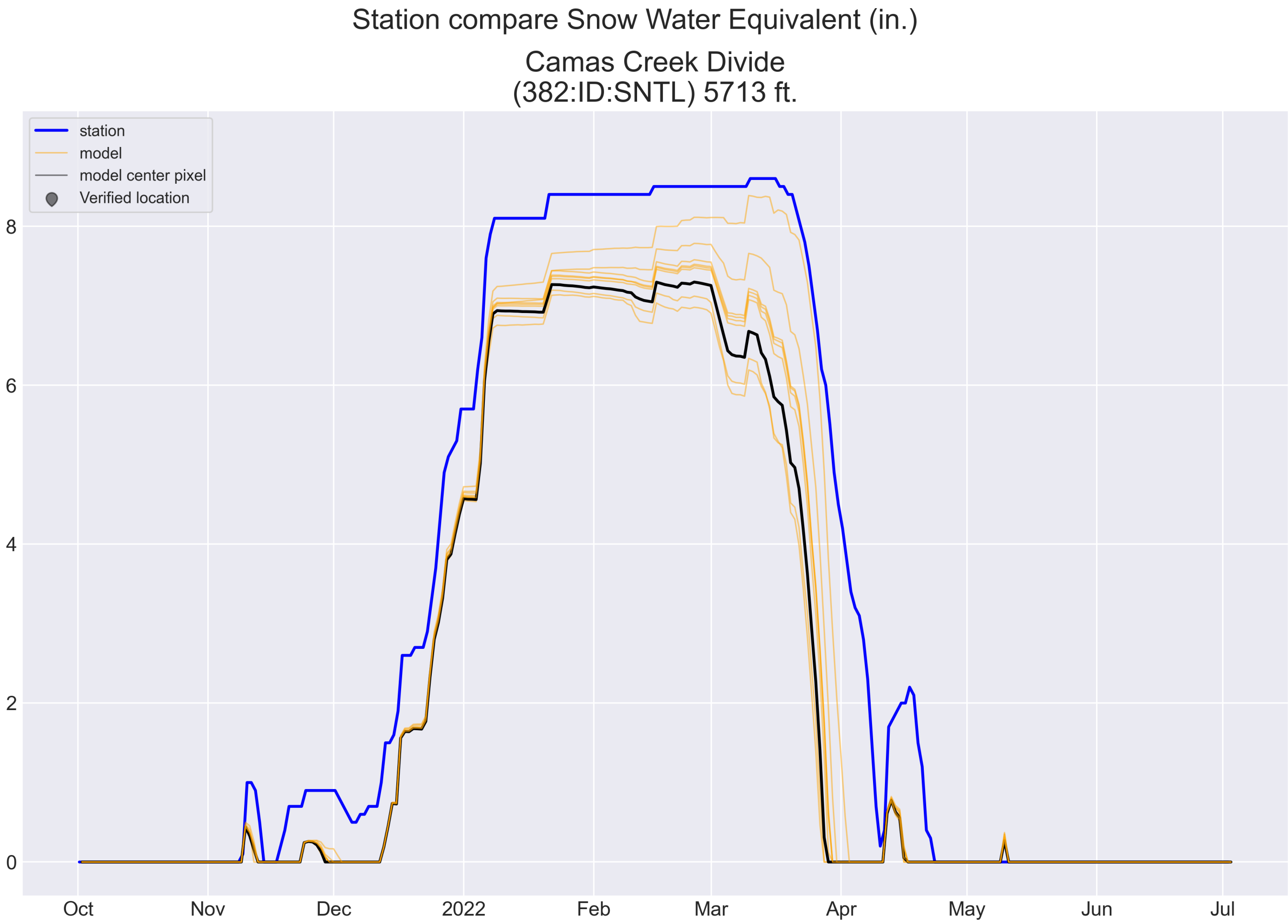

Fig.3 Comparison of modeled and measured Snow Water Equivalent [in.]. Comparisons of model pixels to measured station data are performed with the 9 nearest model pixels to give a more spatially complete view of model behavior.

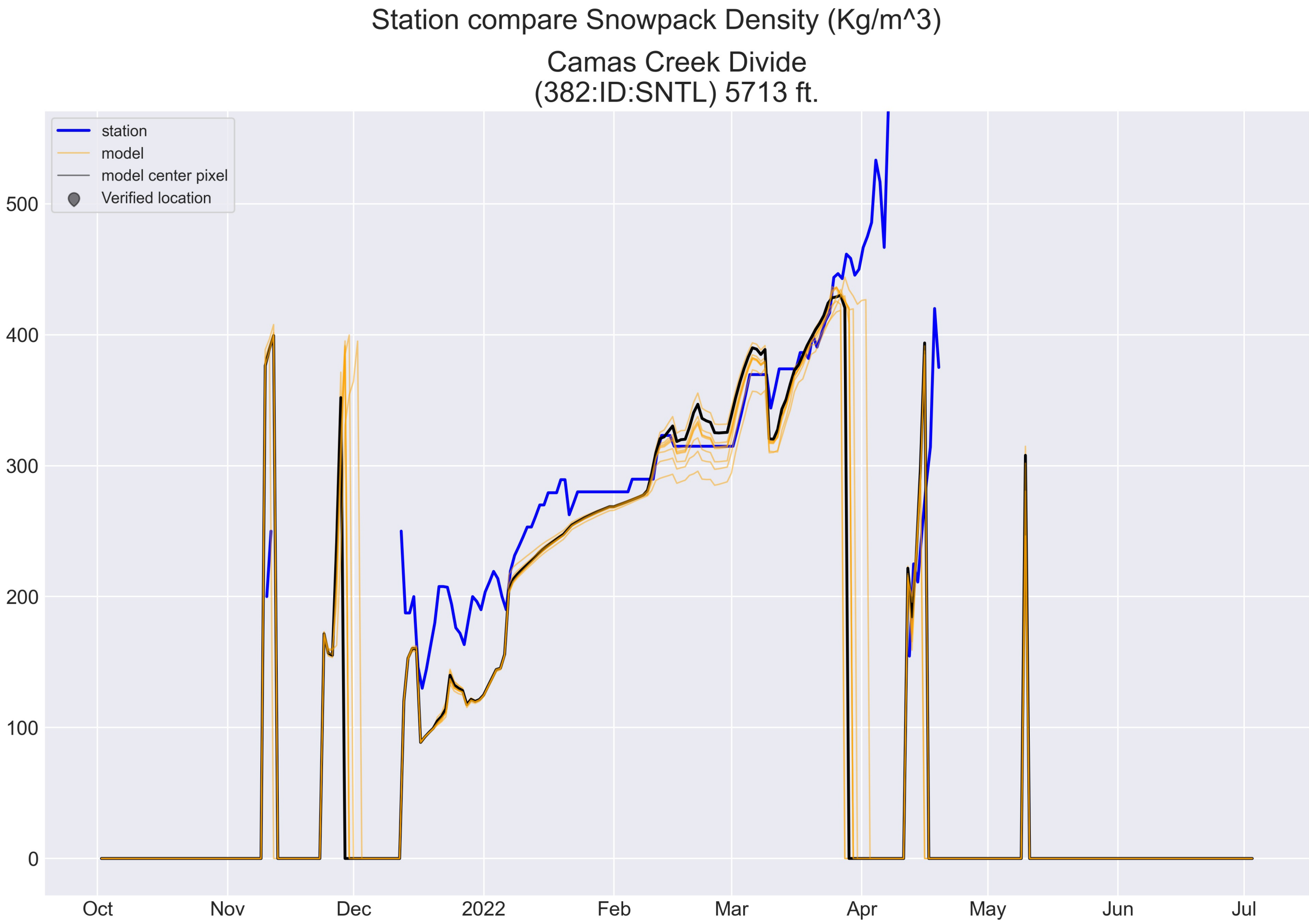

Fig.4 Comparison of modeled and measured Snowpack Density [Kg/m^3]. Comparisons of model pixels to measured station data are performed with the 9 nearest model pixels to give a more spatially complete view of model behavior.

Model Description

Modeled data is generated using the M3 Works model framework BOREAS, which is the next generation of the Automated Water Supply Model (AWSM). Underlying forcing data is sourced from the High Resolution Rapid Refresh model (HRRR). BOREAS simulates the snowpack state using the physically based, distributed energy and mass balance snow model, iSnobal.

This report is brought to you by the team at M3 Works: Mark Robertson, Micah Johnson, Micah Sandusky, Christina Aragon, and Andrew Slaughter.

More information about M3 Works can be found at m3works.io.

Additional Details

Total Snow Water Equivalent values (TAF) on 2022-07-02

ELEVATION RANGE	TOTAL

Accumulated Total Surface Water Input values (TAF) from 2022-07-01 to 2022-07-02

ELEVATION RANGE	TOTAL

Change in Total Snow Water Equivalent values (TAF) from 2022-07-01 to 2022-07-02

ELEVATION RANGE	TOTAL